# Customer Churn & Retention Analysis

Insights Report | Power BI | Telco Customer Churn Dataset

## 1. Project Overview

This project explores customer churn in a telecommunications dataset using Power BI. The dataset contains 7,043 customer records and includes information about customer services, account details, and churn status. The purpose is to summarize churn and retention and examine how churn patterns vary across selected service categories.

## 2. Key Performance Indicators

| Metric | Value | Interpretation |
|---|---|---|
| Total customers | 7,043 | Total customer records in the dataset. |
| Churned customers | 1,869 | Customers marked as having left. |
| Retained customers | 5,174 | Customers not marked as churned. |
| Churn rate | 26.54% | Approximately 27 out of every 100 customers churned. |
| Retention rate | 73.46% | Approximately 73 out of every 100 customers were retained. |

## 3. Analysis Areas

**Internet Service:** Compare churn rates across internet service categories to identify segments that may need closer investigation.

**Tech Support:** Compare customers with and without technical support. Differences may help the business decide whether support access or service experience warrants further review.

**Online Security:** Compare churn across online security subscription groups to understand whether retention patterns differ by service adoption.

## 4. Insights and Business Interpretation

**Overall churn:** The reported churn rate of 26.54% means that slightly more than one quarter of the customer base is marked as churned. This makes churn monitoring an important business priority.

**Retention opportunity:** With 5,174 customers retained, the business has a substantial active customer base. Understanding what distinguishes retained customers from churned customers may help shape targeted retention actions.

**Service-level comparisons:** Internet service, tech support, and online security are useful dimensions for segmentation. The specific highest- and lowest-churn groups should be stated only after checking the actual visual values in the Power BI report.

## 5. Recommendations

• Review churn rates by customer segment and prioritize groups with consistently higher churn.

• Collect customer feedback to understand service issues, billing concerns, and unmet needs.

• Evaluate whether clearer onboarding and easier access to technical support improve the customer experience.

• Consider targeted retention offers for at-risk segments and measure their results before scaling.

• Track churn rate and retention rate over time to assess whether interventions are effective.

## 6. Limitations

This report summarizes dataset-level KPIs and proposed analysis areas. It does not claim that any service feature causes churn. Service-level findings should be confirmed against the actual Power BI visuals, and recommendations should be validated with further analysis or customer feedback.

## 7. Conclusion

The project demonstrates how Power BI can summarize customer churn, compare customer segments, and communicate business-focused retention opportunities. The dashboard can support investigation and decision-making when paired with verified segment-level findings and ongoing monitoring.